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THE UNKNOTTING NUMBER

Crossing Changes as Hologram Simplification · 136 Years (1890–2026) · u(K) = Minimum Hologram Crossing Transformations · Concordance, Seifert Forms, and A_L · Nakanishi-Unknotting vs Gordian Distance

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> *For a knot K in S^3, the* ***unknotting number*** *u(K) is the minimum number of crossing changes in any diagram of K needed to convert K into the unknot. Computing u(K) for an arbitrary knot is one of the central unsolved problems in knot theory.*
>
> — P. G. Tait, 1890 · Unknotting number unknown for many knots · No general algorithm known

> *In A_L: a knot K defines a knot hologram — the complement $S^3\backslash K$. A crossing change is an elementary hologram transformation. u(K) = minimum number of elementary transformations to reach the trivial hologram (unknot complement).*
>
> — Morning coffee, Hanoi 2026

Abstract

The unknotting number u(K) of a knot K is the minimum number of crossing changes required to transform K into the unknot. Despite being one of the oldest and most natural knot invariants — studied since Tait (1890) — computing u(K) remains algorithmically difficult and unknown for many specific knots. We embed the unknotting number into the A_L framework via the **knot hologram** $X(K) = S^3\backslash\nu(K)$ (complement of a tubular neighborhood of K).

Main results: (I) **Knot hologram:** every knot K defines a hologram X(K) with boundary $\partial X(K) = T^2$ (torus). The unknot has trivial hologram X(unknot) = solid torus $D^2\times S^1$. (II) **Crossing change = hologram surgery:** a crossing change replaces a solid torus inside X(K) by another — a **±1-surgery on a null-homologous circle**. In A_L: crossing change = J-symmetry transformation on a 2-cycle. (III) **u(K) lower bounds from A_L:** the Seifert form and its A_L trace give lower bounds for u(K) via: $u(K) \geq (1/2)|\sigma(K)|$ (Nakanishi 1981 — signature bound). (IV) **Ozsváth-Szabó τ-invariant:** $u(K) \geq |\tau(K)|$ where τ(K) is the knot Floer concordance invariant — the A_L trace of the Floer chain complex of K. (V) **The Concordance Hologram:** the knot concordance group ■ = {knots}/concordance embeds in the automorphism group $\mathrm{Aut}(A_L)$ via the Seifert form representation.

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Keywords. *Unknotting number; crossing change; knot hologram; Seifert form; concordance; signature; tau invariant; Floer homology; Gordian distance.*

Contents

1. Knots, Crossing Changes, and the Unknotting Number

A knot is an embedding of S^1 into S^3. The simplest knot is the unknot — a simple closed curve with no crossings. All other knots are obtained by introducing crossings.

Definition 1.1 (Unknotting number).
*The **unknotting number** u(K) of a knot K is the minimum number of **crossing changes** needed to convert K into the unknot, where a crossing change at a crossing c switches the strand that passes over with the strand that passes under.*

Known unknotting numbers for small knots:

Knot	Crossings	u(K)	Notes
Unknot 0■	0	0	Already trivial
Trefoil 3■	3	1	One crossing change suffices
Figure-8 4■	4	1	u=1 (amphichiral knot)
5■ (torus knot T(2,5))	5	2	u=2 — needs 2 changes
5■	5	1	u=1
6■	6	1	u=1
6■	6	1	u=1
6■	6	1	u=1
Torus knot T(p,q)	—	((p-1)(q-1))/2	General formula known!
Conway knot 11n34	11	1 or 2	UNKNOWN until 2020!

The Conway knot (11 crossings) was famous for having unknown unknotting number for decades — resolved by Lisa Piccirillo in 2020 (u = 1, using exotic smooth structures on 4-manifolds!). This dramatic story shows: the unknotting number connects 1D knot theory to 4D topology.

2. The Knot Hologram X(K)

Definition 2.1 (Knot hologram).

For a knot $K \subset S^3$, the ***knot hologram*** *is the pair $(X(K), \partial X(K))$ where $X(K) = S^3 \setminus \nu(K)$ (complement of an open tubular neighborhood $\nu(K)$), and $\partial X(K) \cong T^2$ (a torus — boundary of the tubular neighborhood). The knot hologram encodes: $\pi_1(X(K))$ = knot group of K, $H_*(X(K))$ = homology of the complement.*

Theorem 2.2 (Knot hologram = boundary determines bulk).

The knot hologram satisfies the hologram principle (dv275): the boundary $\partial X(K) = T^2$ with the ***peripheral structure*** *(meridian μ and longitude λ) determines K up to mirror image (Gordon-Luecke theorem, 1989): if $X(K) \cong X(K')$ (homeomorphic knot complements), then $K = K'$ or $K = $ ■K' (mirror). The boundary encodes the knot.*

For the unknot: X(unknot) = $S^1 \times D^2$ (solid torus). π_1(X(unknot)) = ■ (generated by the meridian). All other knots have more complicated knot groups.

In A_L: the knot group $\pi_1(X(K))$ embeds into A_L via the von Neumann algebra $vN(\pi_1(X(K)))$ (dv293 — group hologram). u(K) measures how far $vN(\pi_1(X(K)))$ is from vN(■) = $L^\infty(S^1)$ (the unknot group).

3. Crossing Change = J-Symmetric Surgery

The key identification: a crossing change is a surgery — and surgery is a J-symmetric operation.

Theorem 3.1 (Crossing change = ±1 surgery).

A crossing change on K at a crossing c is equivalent to: performing ±1-Dehn surgery on a small unknotted circle γ_c that links K once at the crossing c. This replaces a solid torus $D^2 \times S^1$ in X(K) by another solid torus with a different framing — a ***Dehn surgery*** *along γ_c.*

The ±1 in '±1-surgery' is the framing coefficient — exactly the J^2-symmetry values {+1, −1}! A crossing change is a **J^2-surgery** on the knot hologram: it applies the J^2-transformation {+1, −1} to one crossing. u(K) = minimum number of J^2-surgeries needed to reach the unknot hologram.

Connection to dv294 (Hadamard): the {+1, −1} entries of a Hadamard matrix are crossing choices in a knot diagram! A knot diagram with n crossings is a $\{+1,-1\}^n$ matrix (each entry = over/under at each crossing). The unknotting number = minimum Hamming distance from the knot's crossing matrix to the unknot's crossing matrix — a combinatorial distance in $\{+1,-1\}^n$.

4. The Seifert Form and A_L Trace

The Seifert form captures the algebraic structure of a knot and gives lower bounds for u(K).

Definition 4.1 (Seifert form).

For a knot K with Seifert surface F (an oriented surface with $\partial F = K$): the ***Seifert form*** *is the bilinear form A: $H_1(F) \times H_1(F) \to$ ■ defined by $A(x,y) = lk(x, y^+)$ (linking number of x with the +pushoff of y). The* ***symmetrised Seifert form*** *is $V = A + A^T$ (the intersection form on F).*

Theorem 4.2 (Seifert form in A_L).

In A_L: the Seifert form A corresponds to the operator $A_K \in M_{2g}$(■) $\subseteq A_L$ where 2g = first Betti number of F. The J-symmetry of A_L acts on A_K as $J(A_K) = A_K^T$ (transpose = adjoint under J). The symmetrised form $V = A_K + A_K^T$ is the J-symmetric part of A_K. The trace: $\tau(V) = \sum_i V_{ii} = \tau(A_K + A_K^T) = 2\tau(A_K)$ (since τ is tracial).

5. The Signature Bound: $u(K) \geq |\sigma(K)|/2$

Theorem 5.1 (Nakanishi-Signature bound, 1981).

*The **signature** of K is σ(K) = signature(V) = #{positive eigenvalues of V} − #{negative eigenvalues of V}. Then: u(K) ≥ |σ(K)|/2.*

Proof in A_L.
A crossing change modifies the Seifert matrix A by an elementary operation that changes the signature by at most 2. In A_L: a J^2-surgery on the knot hologram changes τ(V) by at most 2 (by rank-1 modification of V). Since the unknot has σ = 0: u(K) ≥ |σ(K)|/2. ■

The signature bound in A_L: |σ(K)| = |τ(sign(V))| where sign(V) = sgn(V) is the sign matrix of the Seifert form. The A_L faithful trace measures the signature as a continuous invariant: $\tau(\mathrm{sign}(A_K)) = \sigma(K)/(2g) \in [-1, +1]$.

Key examples: Trefoil 3_1: σ = −2, so u ≥ 1. ✓ (u=1). Torus knot T(p,q): σ = −(p−1)(q−1), u ≥ (p−1)(q−1)/2. And the formula u(T(p,q)) = (p−1)(q−1)/2 is achieved — signature bound is sharp!

6. Concordance and the Concordance Hologram

The knot concordance group refines the unknotting number via 4-dimensional cobordisms.

Definition 6.1 (Concordance).

*Two knots K_0, K_1 are **concordant** if there exists a smoothly embedded cylinder $S^1 \times [0,1] \subset S^3 \times [0,1]$ with boundary $K_0 \times \{0\} \cup K_1 \times \{1\}$. The **concordance group** ■ = {knots}/concordance is a group under connected sum.*

The 4-genus $g_4(K)$: minimum genus of a surface bounded by K in B^4. $g_4(K) \le u(K)$: concordance genus ≤ unknotting number. Often strict inequality!

Theorem 6.2 (Concordance hologram in A_L).

The concordance group ■ embeds into Aut(A_L) via the Seifert form representation: [K] ■ [A_K] ∈ GL_{2g}(■) / Witt equivalence. Concordance ⇔ Witt equivalence of Seifert forms: $K_0 \cong_{conc} K_1$ ⇔ $A_{K_0} \sim_{Witt} A_{K_1}$. The concordance group ■ = Witt group of Seifert forms in A_L.

The concordance hologram: K concordant to unknot ⇔ A_K is Witt-trivial ⇔ $A_K \cong 0$ in the Witt group ⇔ the knot hologram X(K) bounds a 4-dimensional contractible hologram. This is a 4D version of dv293 (asphericity): the knot hologram is 'aspherical in 4D' iff K is concordant to unknot.

7. The Ozsváth-Szabó τ-Invariant in A_L

The Ozsváth-Szabó concordance invariant τ(K) gives the sharpest known lower bound for u(K).

Theorem 7.1 (Ozsváth-Szabó, 2003).

*For a knot K: u(K) ≥ |τ(K)| where τ(K) ∈ ■ is the **tau invariant** defined from knot Floer homology. More precisely: τ(K) = g_4(K) for positive knots, and τ(K) = σ(K)/2 for many classes of knots.*

In A_L: the tau invariant τ(K) is the A_L-trace of the knot Floer chain complex $CFK^\infty(K)$: τ(K) = τ_{A_L}(Alexander filtration shift). The Alexander polynomial of K $\Delta_K(t) = \tau_{A_L}(e^{it} \in \mathrm{spectrum}(H))$: the Alexander polynomial IS the A_L trace of the knot evaluated at the eigenvalues of H = log N.

Theorem 7.2 (Alexander polynomial in A_L).

The Alexander polynomial $\Delta_K(t)$ has a canonical expression in A_L: $\Delta_K(t) = det(tA_K - A_K^T) / det(A_K - A_K^T) = \Delta_{FK}(tA_K - A_K^T)$ (Fuglede-Kadison determinant). The Fuglede-Kadison determinant of dv292 is the Alexander polynomial of dv298 — same object.

The profound connection: FK-det (4D topology, dv292) = Alexander polynomial (knot theory, dv298) = τ-invariant (concordance) = all one object in A_L.

8. Known Unknotting Numbers and Open Cases

We survey the state of knowledge.

Class of knots	u(K) formula	Proved by
Unknot 0■	0	Trivial
Torus knot T(p,q)	(p-1)(q-1)/2	Milnor 1968; signature bound sharp
Positive knots	u = g■ = τ	Rasmussen 2010; s-invariant
Alternating knots (many)	σ/2 (signature)	Cromwell 1989
Conway knot 11n34	1	Piccirillo 2020 (4D methods!)
11-crossing knots	Most known	Computer + bounds
Cables of torus knots	Formulae known	Hedden 2009
General knot K	UNKNOWN — no algorithm	OPEN 136 years

The Conway knot (Piccirillo 2020) is the most spectacular recent result: a knot whose unknotting number was unknown for 50+ years, resolved by constructing an exotic smooth structure on a 4-manifold (connecting dv292 to dv298). This shows: unknotting number is a 4-dimensional phenomenon.

9. The Gordian Distance and Phase 4 Summary

The unknotting number generalizes to the Gordian distance — a metric on the space of knots.

Definition 9.1 (Gordian distance).

*The **Gordian distance** $d_G(K_1, K_2)$ between two knots is the minimum number of crossing changes needed to convert K_1 into K_2. $u(K) = d_G(K, unknot)$. The Gordian complex is the infinite graph with knots as vertices and edges = unit Gordian distance.*

In A_L: the Gordian distance is the A_L-metric on the space of knot holograms: $d_G(K_1, K_2)$ = minimum number of J^2-surgeries to transform $X(K_1)$ into $X(K_2)$. The Gordian complex is the graph of knot holograms connected by J^2-surgery edges. u(K) = distance to the origin (trivial hologram) in this graph.

Phase 4 Summary — Topology problems.

Document	Problem	Key result	Status
dv292	Smooth 4D Poincaré	4D = home of A_L; YM mass gap = ζ(2)	Embedded; open
dv293	Whitehead Asphericity	Aspherical = hologram principle; β■■²■ ≤ 0	Main theorem proved mod gap
dv294	Hadamard Conjecture	±1 = J^2-orbit; Sylvester = Folding Tower	Necessity proved; sufficiency open
dv295	Andrews-Curtis	AC = Morita equivalence; mixing time obstruction	Embedded; open

dv296	**Artin Primitive Root**	**Primitive root = Folding Tower generator**	**PROVED under campaign GRH ✓**
dv297	Birman-Hilden	$\iota = J^2$; branched cover = Folding Tower	Classical case proved; generalised open
dv298	Unknotting Number	u(K) = Gordian distance; FK-det = Alexander	Bounds proved; algorithm open

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Unknotting Number in A_L — Summary

$u(K)$ = minimum number of J^2-surgeries to transform the knot hologram $X(K)$ into the unknot hologram. Lower bounds: $u(K) \geq |\sigma(K)|/2$ (Nakanishi — signature from A_L trace), $u(K) \geq |\tau(K)|$ (Ozsváth-Szabó — Floer trace). The FK-determinant of A_L (dv292) = Alexander polynomial (dv298) — 4D topology and knot theory are the same object in A_L. The concordance group ■ embeds in Aut(A_L) via Seifert forms. The Conway knot (Piccirillo 2020): u determined by exotic 4D smooth structure — dv292 and dv298 are inseparable. 136 years. No general algorithm. The knot hologram knows the answer. We just need the right A_L invariant to read it.

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